AGATHE TOURNANT

Biomedical Engineer MSc. · Life Sciences Consultant

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PROFILE

Biomedical engineer MSc. with a strong foundation in deep learning for neural data, combining a Master's from ETH Zurich and MIT research experience in interpretable AI for computational psychiatry. My hands-on experience spans machine learning for neuroimaging, explainable AI, and high-dimension image processing. My current position in life sciences consulting adds a strategic, cross-functional perspective on AI in regulated healthcare environments alongside solid project management skills. Motivated to advance applied machine learning in healthcare, neuroscience, and precision medicine.

SKILLS

ML & Deep Learning: PyTorch, TensorFlow/Keras, scikit-learn; self-attention / Transformers, U-Net, self-supervised learning, gradient reversal, mixed precision training, synthetic data augmentation

Neural & Biomedical Data: fMRI (resting-state & naturalistic), structural MRI, brain lesion segmentation, functional connectivity, MNI-space registration, multi-site neuroimaging

Programming: Python (advanced), Matlab, R, SQL, C++, Git

AI & Tooling: Python CrewAI, CoreML, SPM, 3D Slicer, n8n

Domain Knowledge: Computational neuroscience, clinical AI, precision medicine, data governance, regulatory compliance

Agile & Delivery: SAFe 6 Agilist, Scrum Master PSM I, cross-functional stakeholder management

PROFESSIONAL EXPERIENCE

Data & Strategy Consultant, Life Sciences – Capgemini Invent
Zurich, Apr 2024 – Present

Data & AI Strategy Projects

- Supported pharmaceutical company in defining digital, data and AI strategies; assessing enterprise data and system landscapes to identify critical assets and data product candidates across the R&D lifecycle
- Led data compliance assessment against the IDMP regulatory standard for medicinal product data, conducting 50+ SME interviews to map operational pain points and designing conceptual data models and governance frameworks to enable scalable data products
- Supported the central PMO of a large-scale digital workplace transformation program of a top 10 pharmaceutical company to enable an AI-native workplace, coordinating cross-functional stakeholders and operations in a SAFe setup

AI Capability Building & Industry Engagement

- Contributed to internal initiatives building Capgemini's AI capabilities for early drug development; supported external representation at BioTechX; contributed to content showcased at NVIDIA GTC Paris and strengthened partnerships with key technology partners including NVIDIA
- Developed internal assets for agentic AI applications using CrewAI, designing and orchestrating multi-agent systems to extract insights from large datasets, and n8n, to design automatic workflows for the Invet

AI Medical – Machine Learning Engineer Intern
Zurich, Mar 2023 – Jul 2023

- Optimised and trained U-Net-based deep learning models in Python (PyTorch) for automated detection of brain lesions (multiple sclerosis and brain metastases) in 3D MRI, targeting clinical decision support for neurological disorders
- Designed and implemented a synthetic lesion augmentation pipeline to improve model generalisation on rare and small lesions; evaluated model variants across sensitivity/specificity trade-offs using Dice coefficient stratified by lesion size

- Conducted MNI-space registration, preprocessing, and image signal analysis to optimise normalisation parameters and improve training stability across multi-site datasets; ran model depth and mixed-precision ablation experiments
- Converted trained Keras models to CoreML format for on-device deployment in the clinical software
- Co-authored technical documentation supporting FDA 510(k) submission, translating model architecture and validation methodology into regulatory-grade language

EDUCATION

M.Sc. Biomedical Engineering Zurich/Boston, Sep 2021 – Mar 2024

- ETH Zurich – Bioimaging track with focus on healthcare machine learning application. GPA: 5.57/6
- MIT (Senseable Intelligence Group, McGovern Institute for Brain Research) – Master's thesis: "Interpretable Deep Learning and Naturalistic Stimuli for Connectivity-Based Identification of Mood Disorders". Grade: 5.75/6

B.Sc. Physics Lausanne/Vienna, Sep 2018 – Jun 2021

- EPFL – General Physics track. GPA: 5.07/6
- TU Wien – Exchange semester (3rd year)

SELECTED PROJECTS

Master's Thesis: Interpretable Deep Learning for Psychiatric fMRI Classification (PyTorch, Matlab)
ETHZ/MIT, Sep 2023 – Mar 2024

- Optimized and trained a self-attention and temporal modeling architecture to learn directed functional connectivity graphs from resting-state and naturalistic fMRI time series, targeting mood disorder (depression, anxiety) classification
- Integrated gradient reversal layers to remove site-specific confounds, producing a site-agnostic model robust to multi-site neuroimaging data
- Achieved mean ROC-AUC > 0.7 on depression classification; interpretability analysis identified aberrant subcortical connectivity (cerebellum, thalamus) as discriminative signatures; naturalistic stimuli outperformed resting-state, supporting the emotion dysregulation hypothesis
- Applied nested cross-validation for hyperparameter tuning, reproducible preprocessing pipelines, and explicit overfitting mitigation strategies across large-scale neuroimaging datasets

Hackathon: Agentic AI for Educational Data Analysis (Python, CrewAI, SQL)
Capgemini, Oct 2024

- Designed and implemented a multi-agent CrewAI system with specialised agents to extract policy-relevant insights from the PIRLS international education dataset

Semester Project: Gender-disaggregated ML Tools for ASD Diagnosis (Python, Matlab, SPM)
ETHZ, Apr – May 2022

- Developed sex-disaggregated diagnostic tools for Autism Spectrum Disorder using regression DCM models and ML classification on neuroimaging data

Semester project: Multi-scale imaging in PD mouse model (Immunohistochemistry, Matlab)
ETHZ, Sep 2022 – Jan 2023

- Analyzed the relationship between the spread of alpha-synuclein, structural connectivity alterations and neurodegeneration in a Parkinson’s disease mouse model

LANGUAGES

French (Native) **English** (Fluent, C1–C2) **German** (Intermediate, B1–B2)